A Counterexample to a Cubic Implication for Hyperfields

Abstract

The proposed implication is false for general hyperfields. We construct a countably infinite counterexample whose multiplicative group is infinite cyclic. The construction uses an explicitly defined infinite difference set. All required incidence properties and all hyperfield axioms are verified directly.

1 Statement and conventions

We will prove the following.

Theorem 1. *There exists a hyperfield*

$$\mathbb{H} = \{0\} \sqcup \{t^n : n \in \mathbb{Z}\}, \qquad t^m t^n = t^{m+n},$$

in which every element is its own additive inverse and

$$0 \in 1 \boxplus t \boxplus t^6,$$

but

$$0 \notin 1 \boxplus t^3 \boxplus t^{18} \boxplus t^7 \boxplus t^7 \boxplus t^7.$$

Consequently, the implication in the question fails for

$$x = 1, \qquad y = t, \qquad z = t^6.$$

Here the symbols t^n are pairwise distinct formal symbols, and $1 = t^0$. In particular, the integer 0, when used as an exponent or an incidence point, must be distinguished from the hyperfield element 0.

For subsets A, B of a set equipped with hyperaddition, we use the convention

$$A \boxplus B = \bigcup_{\substack{a \in A \\ b \in B}} (a \boxplus b).$$

Singleton sets are identified with their elements in such expressions.

We use the usual definition of a commutative hyperfield: its nonzero elements form an abelian multiplicative group, zero is multiplicatively absorbing, and hyperaddition is a commutative, associative, nonempty-valued operation with neutral element 0, unique additive inverses, and reversibility

$$c \in a \boxplus b \quad \Longleftrightarrow \quad b \in c \boxplus (-a).$$

Multiplication must distribute over hyperaddition as an equality of sets.

The connection between projective incidence geometry and hyperfields extending the Krasner hyperfield is developed in [1, Sections 3–4]. The construction below, including the verification of the hyperfield axioms, is self-contained.

2 An infinite difference set

Lemma 2. *There exists an infinite set $D \subseteq \mathbb{Z}_{\geq 0}$ such that*

1.
$$D \cap \{0, 1, \ldots, 38\} = \{0, 1, 6, 20, 23, 27\};$$

2. every nonzero integer has a unique expression $d - e$, where $d, e \in D$.

Proof. For a finite set $A \subseteq \mathbb{Z}_{\geq 0}$, put

$$\Delta(A) = \{a - b : a, b \in A,\ a > b\}.$$

Say that A has distinct positive differences if each element of $\Delta(A)$ arises from exactly one ordered pair (a, b).

Begin with

$$A_0 = \{0, 1, 6, 20, 23, 27\}.$$

Its positive differences are

$$1, 3, 4, 5, 6, 7, 14, 17, 19, 20, 21, 22, 23, 26, 27.$$

These are fifteen distinct integers, accounting for all $\binom{6}{2} = 15$ pairs. Thus A_0 has distinct positive differences.

We first describe an extension step. Suppose that A is a finite set containing A_0, that A has distinct positive differences, and that $n \geq 1$ does not belong to $\Delta(A)$. Set

$$M = \max A, \qquad u = 2M + n + 1, \qquad A' = A \cup \{u, u + n\}.$$

All positive differences in A' occur in the following four sets:

$$\Delta(A), \qquad \{n\}, \qquad \{u - a : a \in A\}, \qquad \{u + n - a : a \in A\}.$$

These sets are pairwise disjoint. Indeed, $n \notin \Delta(A)$, while every integer in either of the last two sets is at least

$$u - M = M + n + 1 > \max\{M, n\}.$$

Every element of $\Delta(A)$ is at most M. Finally, an equality

$$u - a = u + n - b$$

would imply $b - a = n$, contrary to $n \notin \Delta(A)$. Within each of the last two sets, distinct elements of A give distinct differences. Therefore A' again has distinct positive differences. Moreover, $n \in \Delta(A')$.

Now define A_n recursively for $n \geq 1$. If $n \in \Delta(A_{n-1})$, let $A_n = A_{n-1}$. Otherwise apply the extension step with $A = A_{n-1}$ and this value of n. Set

$$D = \bigcup_{n \geq 0} A_n.$$

Every positive integer n belongs to $\Delta(A_n)$, hence is a difference of two elements of D. Such a representation is unique: any two competing representations would have all their entries in some finite A_N, contradicting the distinct-difference property of A_N. Reversing the ordered pair proves existence and uniqueness for negative integers as well.

At every extension step $M \geq 27$, so

$$u = 2M + n + 1 \geq 56 > 38.$$

Thus no new element at most 38 is ever introduced, proving the first assertion. Finally, D is infinite, since a finite set has only finitely many differences. □

Lemma 3. *For $k \in \mathbb{Z}$, define*

$$L_k = D + k = \{d + k : d \in D\}.$$

Regard the integers as points and the sets L_k as lines. Then:

1. *every two distinct points lie on exactly one line;*
2. *every two distinct lines meet in exactly one point;*
3. *every line is infinite;*
4. *integer translations preserve the collection of lines.*

Proof. Let $p \neq q$ be integers. By Lemma 2, there is a unique ordered pair $d, e \in D$ satisfying

$$p - q = d - e.$$

Then

$$k = p - d = q - e$$

is the unique integer for which $p, q \in L_k$. This proves the first assertion.

For $k \neq \ell$, a point of $L_k \cap L_\ell$ corresponds precisely to an ordered pair $d, e \in D$ satisfying

$$k + d = \ell + e, \qquad \text{equivalently} \qquad d - e = \ell - k.$$

There is exactly one such ordered pair, so the intersection consists of exactly one point. Since each L_k is infinite, this also shows that different indices give different lines.

The third assertion follows from the infinitude of D, and the fourth from

$$L_k + s = L_{k+s}.$$

□

For distinct integers m, n, write $L(m, n)$ for their unique common line.

3 Construction and verification of the hyperfield

Let

$$\mathbb{H} = \{0\} \sqcup \{t^n : n \in \mathbb{Z}\},$$

with multiplication

$$t^m t^n = t^{m+n}, \qquad 0a = a0 = 0.$$

For distinct nonzero elements $a = t^m$ and $b = t^n$, define their line by

$$\Lambda(a, b) = \{t^r : r \in L(m, n)\} \subseteq \mathbb{H}^\times.$$

The incidence properties of Lemma 3 hold equally for these lines in $\mathbb{H}^\times$.

Define hyperaddition by

$$0 \boxplus a = a \boxplus 0 = \{a\}$$

and, for $a, b \neq 0$, by

$$a \boxplus b = \begin{cases} \{0, a\}, & a = b, \\ \Lambda(a, b) \setminus \{a, b\}, & a \neq b. \end{cases} \tag{1}$$

Proposition 4. *These operations make $\mathbb{H}$ a commutative hyperfield. Its additive negation is the identity map.*

Proof. The nonzero elements form an infinite cyclic multiplicative group with identity $1 = t^0$. Zero is absorbing by definition.

Nonempty sums, commutativity, and the neutral element. Every line is infinite, so all sums in (1) are nonempty. Commutativity and the additive neutral-element axiom follow directly from the definition.

Additive inverses and reversibility. For all $a, b \in \mathbb{H}$, the definition gives

$$0 \in a \boxplus b \quad \Longleftrightarrow \quad a = b. \tag{2}$$

Consequently, every element has a unique additive inverse, namely

$$-a = a.$$

To verify reversibility, first suppose that a, b, c are all nonzero. Then $c \in a \boxplus b$ holds precisely when either all three elements coincide, or they are pairwise distinct and lie on a common line. Both conditions are symmetric in a, b, c. If at least one element is zero, membership holds precisely when the other two elements are equal, again a symmetric condition. Hence

$$c \in a \boxplus b \quad \Longleftrightarrow \quad b \in c \boxplus a \quad \Longleftrightarrow \quad b \in c \boxplus (-a),$$

as required.

Associativity. If one of the three elements is zero, associativity follows from the neutral-element rule. We therefore consider nonzero elements.

If all three elements equal a, then

$$(a \boxplus a) \boxplus a = \{0, a\} \boxplus a = \{0, a\}.$$

Suppose next that the multiset of the three elements is $\{a, a, b\}$, where $a \neq b$, and let $\Lambda = \Lambda(a, b)$. Then

$$(a \boxplus a) \boxplus b = \{b\} \cup \big(\Lambda \setminus \{a, b\}\big) = \Lambda \setminus \{a\}.$$

Also, uniqueness of the line through two distinct points gives

$$\begin{aligned}(a \boxplus b) \boxplus a &= \bigcup_{s \in \Lambda \setminus \{a,b\}} \big(\Lambda \setminus \{s, a\}\big) \\ &= \Lambda \setminus \{a\}.\end{aligned}$$

For the last equality, given $v \in \Lambda \setminus \{a\}$, choose $s \in \Lambda \setminus \{a, b, v\}$, which is possible because Λ is infinite.

Now suppose that a, b, c are pairwise distinct and lie on one line Λ. We claim that

$$(a \boxplus b) \boxplus c = \{0\} \cup \Lambda. \tag{3}$$

Every summand on the left is contained in $\{0\} \cup \Lambda$. Choosing the intermediate element $c \in a \boxplus b$ produces both 0 and c. Given $v \in \Lambda \setminus \{c\}$, choose

$$s \in \Lambda \setminus \{a, b, c, v\}.$$

Then $s \in a \boxplus b$ and $v \in s \boxplus c$, proving (3).

Finally, suppose that a, b, c are noncollinear. In this case

$$(a \boxplus b) \boxplus c = \mathbb{H}^\times \setminus \big(\Lambda(a,b) \cup \Lambda(a,c) \cup \Lambda(b,c)\big). \tag{4}$$

To prove the forward inclusion, take

$$s \in \Lambda(a,b) \setminus \{a,b\}.$$

Since $c \notin \Lambda(a,b)$, the elements s, c are distinct. The line $\Lambda(s,c)$ meets $\Lambda(a,b)$ only at s, and meets each of $\Lambda(a,c)$ and $\Lambda(b,c)$ only at c. Therefore

$$s \boxplus c = \Lambda(s,c) \setminus \{s,c\}$$

avoids all three lines on the right of (4), as well as the hyperfield zero.

Conversely, let v lie outside those three lines. The line $\Lambda(c,v)$ meets $\Lambda(a,b)$ in a unique point s. This point is neither a nor b, since otherwise v would belong to $\Lambda(a,c)$ or $\Lambda(b,c)$. Also $v \neq s, c$. Thus

$$s \in a \boxplus b, \qquad v \in s \boxplus c,$$

proving the reverse inclusion.

The formulas just established are independent of the order in which the three elements are listed. Together with commutativity, they imply

$$(a \boxplus b) \boxplus c = a \boxplus (b \boxplus c)$$

in every case. Associativity for subsets follows by taking unions over their elements; hence finite hypersums can be parenthesized freely.

Distributivity. Multiplication by t^k acts on nonzero elements by translating their exponents by k. By Lemma 3, it carries lines to lines, and in particular

$$t^k \Lambda(t^m, t^n) = \Lambda(t^{m+k}, t^{n+k}) \qquad (m \neq n).$$

It follows from (1), including its equal-argument and zero-argument cases, that

$$t^k(a \boxplus b) = t^k a \boxplus t^k b.$$

For multiplication by zero, both sides are $\{0\}$, since $a \boxplus b$ is nonempty. Thus distributivity holds for all scalars. All hyperfield axioms have now been verified. □

4 Failure of the cubic implication

Proof of Theorem 1. Use the hyperfield constructed above and set

$$x = 1, \qquad y = t, \qquad z = t^6.$$

The integers $0, 1, 6$ all belong to D. Consequently,

$$t^6 \in \Lambda(1,t) \setminus \{1,t\} = 1 \boxplus t.$$

It follows that

$$0 \in t^6 \boxplus t^6 \subseteq (1 \boxplus t) \boxplus t^6.$$

Thus the hypothesis in the question is satisfied.

Consider the line

$$\Lambda = \{t^r : r \in D - 20\}.$$

Because $20, 23, 27 \in D$, this line contains

$$1, \qquad t^3, \qquad t^7.$$

On the other hand,

$$t^{18} \notin \Lambda,$$

because $18 \in D - 20$ would imply $38 \in D$, contrary to Lemma 2.

Write

$$u = 1, \qquad v = t^3, \qquad w = t^{18}, \qquad q = t^7, \qquad A = u \boxplus v \boxplus w.$$

The points u, v, w are noncollinear, and $\Lambda(u, v) = \Lambda$. Formula (4) therefore gives

$$A \subseteq \mathbb{H}^\times \setminus \Lambda.$$

In particular,

$$0 \notin A, \qquad q \notin A. \tag{5}$$

The repeated sum of q satisfies

$$q \boxplus q \boxplus q = \{0, q\} \boxplus q = \{0, q\}.$$

Hence

$$\begin{aligned} u \boxplus v \boxplus w \boxplus q \boxplus q \boxplus q &= A \boxplus \{0, q\} \\ &= A \cup \bigcup_{a \in A} (a \boxplus q). \end{aligned}$$

The first set on the last line does not contain zero by (5). By (2), a set $a \boxplus q$ contains zero exactly when $a = q$; this is also excluded by (5). Therefore

$$0 \notin 1 \boxplus t^3 \boxplus t^{18} \boxplus t^7 \boxplus t^7 \boxplus t^7.$$

Finally, multiplication and negation in this hyperfield give

$$x^3 = 1, \qquad y^3 = t^3, \qquad z^3 = t^{18}, \qquad -xyz = xyz = t^7.$$

Thus the last displayed nonmembership is precisely the failure of the conclusion asked about in the question. □

References

[1] A. Connes and C. Consani, *The hyperring of adèle classes*, Journal of Number Theory **131** (2011), no. 2, 159–194.